

ABSTRACTION AND INTERFACES

Q1. What is abstraction?

- a) Hiding data
- b) Hiding implementation details and showing functionality
- c) Showing all details
- d) Using classes

Answer: b

Q2. How is abstraction achieved in Java?

- a) Classes only
- b) Abstract classes and interfaces
- c) Constructors
- d) Loops

Answer: b

Q3. Which keyword is used to declare an abstract class?

- a) interface
- b) abstract
- c) class
- d) extends

Answer: b

Q4. Can we create object of abstract class?

- a) Yes
- b) No
- c) Only once
- d) Only static

Answer: b

Q5. What is an abstract method?

- a) Method with body
- b) Method without body
- c) Static method
- d) Final method

Answer: b

Q6. If a class has an abstract method, it must be?

- a) final
- b) static
- c) abstract
- d) public

Answer: c

Q7. Can abstract class have constructors?

- a) Yes
- b) No
- c) Only private
- d) Only static

Answer: a

Q8. Can abstract class have non-abstract methods?

- a) Yes
- b) No
- c) Only static
- d) Only final

Answer: a

Q9. Which keyword is used to implement an interface?

- a) extends
- b) implements
- c) inherits
- d) interface

Answer: b

Q10. Can a class implement multiple interfaces?

- a) Yes
- b) No
- c) Only two
- d) Only one

Answer: a

Q11. What is default access modifier of interface methods?

- a) private
- b) protected
- c) public
- d) default

Answer: c

Q12. Variables in interface are by default?

- a) private
- b) protected
- c) public static final
- d) static only

Answer: c

Q13. Can interface have constructors?

- a) Yes
- b) No
- c) Only default
- d) Only parameterized

Answer: b

Q14. What is the difference between abstract class and interface?

- a) Abstract class supports multiple inheritance
- b) Interface supports multiple inheritance
- c) Both same
- d) None

Answer: b

Q15. What will be output?

```
abstract class A {  
    abstract void show();  
}  
class B extends A {
```

```
void show() {  
    System.out.println("Hello");  
}  
}  
public class Test {  
    public static void main(String[] args) {  
        A obj = new B();  
        obj.show();  
    }  
}
```

- a) Error
- b) Hello
- c) Null
- d) Nothing

Answer: b

Q16. What happens if subclass does not implement abstract method?

- a) Compiles
- b) Runtime error
- c) Must be declared abstract
- d) JVM ignores

Answer: c

Q17. Can abstract class be final?

- a) Yes
- b) No
- c) Only static
- d) Only private

Answer: b

Q18. What will be output?

```
interface A {  
    void show();  
}  
class B implements A {  
    public void show() {  
        System.out.println("Hi");  
    }  
}  
public class Test {  
    public static void main(String[] args) {  
        A obj = new B();  
        obj.show();  
    }  
}
```

- a) Error
- b) Hi
- c) Null
- d) Nothing

Answer: b

Q19. Can interface extend another interface?

- a) Yes
- b) No
- c) Only once
- d) Only abstract

Answer: a

Q20. What will be output?

```
interface A {  
    int x = 10;  
}  
  
public class Test {  
    public static void main(String[] args) {  
        System.out.println(A.x);  
    }  
}
```

- a) 0
- b) 10
- c) Error
- d) Null

Answer: b

Q21. Can we override default methods in interface?

- a) Yes
- b) No
- c) Only static
- d) Only private

Answer: a

Q22. What is functional interface?

- a) Interface with many methods
- b) Interface with one abstract method
- c) Interface with no methods
- d) Class

Answer: b

Q23. Which annotation is used for functional interface?

- a) @Override
- b) @FunctionalInterface
- c) @Interface
- d) @Abstract

Answer: b

Q24. What will be output?

```
abstract class A {  
    A() {  
        System.out.println("A");  
    }  
}  
  
class B extends A {  
    B() {  
        System.out.println("B");  
    }  
}
```

```
public class Test {  
    public static void main(String[] args) {  
        new B();  
    }  
}
```

- a) A B
- b) B A
- c) A
- d) Error

Answer: a

Q25. Can interface have static methods?

- a) Yes
- b) No
- c) Only private
- d) Only final

Answer: a

Q26. What will be output?

```
interface A {  
    default void show() {  
        System.out.println("A");  
    }  
}  
  
class B implements A {  
}  
  
public class Test {  
    public static void main(String[] args) {  
        B obj = new B();  
        obj.show();  
    }  
}
```

- a) A
- b) Error
- c) Null
- d) Nothing

Answer: a

Q27. Can interface methods be private?

- a) Yes (Java 9+)
- b) No
- c) Only public
- d) Only protected

Answer: a

Q28. What will happen?

```
interface A {  
    void show();  
}  
  
class B implements A {  
}
```

- a) Compiles
- b) Runtime error

- c) Compile-time error
- d) No output

Answer: c

Q29. Which is better for full abstraction?

- a) Abstract class
- b) Interface
- c) Class
- d) Method

Answer: b

Q30. What will be output?

```
interface A {  
    default void show() {  
        System.out.println("A");  
    }  
}  
interface B {  
    default void show() {  
        System.out.println("B");  
    }  
}  
class C implements A, B {  
    public void show() {  
        A.super.show();  
    }  
}  
public class Test {  
    public static void main(String[] args) {  
        new C().show();  
    }  
}
```

- a) A
- b) B
- c) Error
- d) A B

Answer: a